OBJECTS OF FINITE LENGTH AND FILTRATIONS WITH SIMPLE SUCCESSIVE QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument in both directions; the base case of the induction is stated and justified in one sentence, the induction step names the two lemmas it uses and the short exact sequence it applies them to, and no step of the proof is left to the reader.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Artinian and Noetherian objects. Same chapter, the definitions of an Artinian and of a Noetherian object. Supply the two chain conditions that appear in the statement and are used to make the chain of subobjects exist and terminate.
- Chain conditions and short exact sequences. Same chapter, Lemma lemma-ses-artinian and Lemma lemma-ses-noetherian. Supply that in a short exact sequence the middle object satisfies a chain condition as soon as the two outer objects do, which is the induction step.
- Simple objects. Same chapter, the definition of a simple object. Supplies the notion used to describe the successive quotients of the filtration.

Source reference. Original lemma 0FCH: sources/stacks-homology/pinned-original-source.tex, source lines 2207–2207. Navigation pointer invoked at proof line 2255; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0FCI: sources/stacks-homology/pinned-original-source.tex, source lines 2218–2218. Navigation pointer invoked at proof line 2255;

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the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$. The following are equivalent*

(1) *A is Artinian and Noetherian, and*
(2) *there exists a filtration $0 \subset A_1 \subset A_2 \subset \ldots \subset A_n = A$ by subobjects such that A_i/A_{i-1} is simple for $i = 1, \ldots, n$.*

Proof. Assume (1). If A is zero, then (2) holds. If A is not zero, then there exists a smallest nonzero object $A_1 \subset A$ by the Artinian property. Of course A_1 is simple. If $A_1 = A$, then we are done. If not, then we can find $A_1 \subset A_2 \subset A$ minimal with $A_2 \neq A_1$. Then A_2/A_1 is simple. Continuing in this way, we can find a sequence $0 \subset A_1 \subset A_2 \subset \ldots$ of subobjects of A such that A_i/A_{i-1} is simple. Since A is Noetherian, we conclude that the process stops. Hence (2) follows.

Assume (2). We will prove (1) by induction on n. If $n = 1$, then A is simple and clearly Noetherian and Artinian. If the result holds for $n-1$, then we use the short exact sequence $0 \to A_{n-1} \to A_n \to A_n/A_{n-1} \to 0$ and Lemmas 0FCH and 0FCI to conclude for n. □